Somaiya Vidyavihar University
K J Somaiya School of Engineering

Batch: **Roll No.:**
Experiment / assignment / tutorial No. 02
Grade: AA / AB / BB / BC / CC / CD /DD
Signature of the Staff In-charge with date

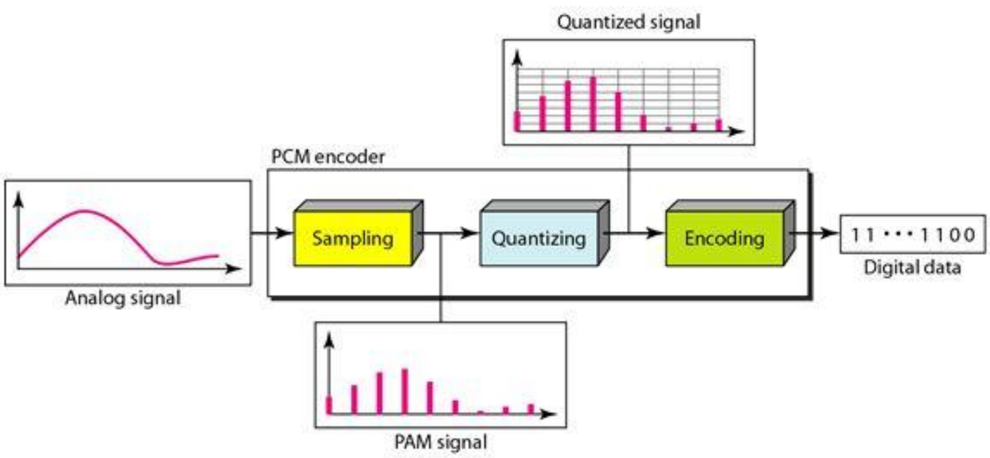
Title: Analog to Digital Conversion

Objectives:

1. **To understand the steps in analog to digital conversion with given example**
2. **To simulate the analog to digital conversion steps in Simulink and analyze the results**

OUTCOME: apply concept of sampling for reliable communication

Theory:



Problem Statement:

- Assuming that a 3-bit unipolar ADC channel accepts analog input (sine wave) ranging from 1 to -1 volts with frequency of 2 Hz. Write a program to plot sampled signal, quantized signal, and encoded signal. Show all background mathematical calculations.
- Design a model in Simulink for the above statement.

Expected Output:

- Various calculations to design Simulink model
- Plot of sampled, quantized and encoded signal

Activity

- **Watch the following video and answer the questions below:**
  - **https://www.youtube.com/watch?v=I1AhW6_HQpI**
  - Write reflection of video in not more than 8 to 10 lines.
  - According to you which system is better? Analog or Digital?

Discussion/Conclusion

Signature of faculty in-charge